

# Sex-biased dispersal and growth in sablefish (*Anoplopoma fimbria*) in the northeastern Pacific Ocean

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**Abstract** To examine the potential linkage between sex-biased growth and dispersal in demersal fish, we studied the movement distance and growth of tag-recaptured sablefish (*Anoplopoma fimbria*). Tagging was conducted by teams from Japan and the United States. From 1980 to 1984, 58,918 sablefish were captured by Japanese vessels and tagged with anchor tags, and between 1980 and 2007, 3319 sablefish were recaptured in the Bering Sea and northeastern Pacific. Of these 3319 fish, we analyzed data on 659 individuals for which we have reliable data on recapture location, sex and fork length at the time of release and recapture, and elapsed time longer than 10 days. Annual growth increments (mean±S.D.) were  $2.5\pm6.5$  cm (males) and  $3.5\pm8.6$  cm (females), and the movement distances (assumed to be straight) were  $712\pm800$  km (males) and  $877\pm920$  km (females). Females grew larger and moved longer distances than males. We calculated the growth increment deviation that accounted for the effects of elapsed time, curvilinear growth, and initial fork length at release using a von Bertalanffy growth model. The growth increment deviation increased significantly with the distance moved over elapsed time. In addition, the relationship between distance moved over elapsed

time and the growth deviation differed significantly between sexes; females grew faster than males at a given movement distance. We suggest that female-biased dispersal is a factor generating female-biased growth in sablefish.

**Keywords** Bering Sea · Demersal fish · Fecundity-selection · Movement · Sexual selection · Sexual size dimorphism

## Introduction

Sexual size dimorphism occurs often in animals. For example, mature males are generally larger than mature females in mammals (Ruckstuhl and Neuhaus 2002), whereas mature females are generally larger than mature males in fishes, especially in demersal fishes (e.g., Itaya and Fujioka 2006; Ito et al. 2008; Narimatsu et al. 2008). Stearns (1992) suggested that sexual selection strongly favors larger male size in polygynous mammals, whereas fecundity selection favors larger females in fishes, because fish fecundity increased isometrically with body weight. However, male-biased body sizes have been reported in some fish species that show intense competition between males for females (Beacham and Murray 1985; Blair et al. 1993; Katano 1998; Quinn et al. 2001). In addition, patterns of sexual size dimorphism sometimes vary among populations within fish species, and the variation in the relative strengths of sexual-

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selection and fecundity-selection would produce the observed sexual size dimorphism pattern (Holtby and Healey 1990; Tamate and Maekawa 2006).

Sex-biased dispersal also occurs in many animals, including fishes (Greenwood 1980; Dobson 1982; Clarke et al. 1997; Hendry et al. 2004). It has been suggested that sex-biased dispersal is related to mating systems and territorial behavior (Greenwood 1983). Male-biased movement has been reported in some fishes, e.g., great white shark, *Carcharodon carcharias* (Pardini et al. 2001), and brook trout, *Salvelinus fontinalis* (Hutchings and Gerber 2002). By contrast, female-biased movement has been reported in partially migratory salmonids, in which migrants are larger than residents, i.e., migrants move to productive habitats to feed and attain larger body size (Jonsson and Jonsson 1993; Tamate and Maekawa 2004; Koizumi et al. 2006). In addition, the movements of female hammerhead shark, *Sphyrna lewini*, to a pelagic habitat resulted in increased feeding success and rapid growth compared with males (Kimley 1987). Although not all demersal fishes show long distant movement, if it occurs, it is postulated that female-biased movements may be a cause of the female-biased sexual size dimorphisms reported for many demersal fishes (Andrews et al. 1999; Itaya and Fujioka 2006; Ito et al. 2008; Narimatsu et al. 2008). In the case of sablefish, *Anoplopoma fimbria*, both female-biased sexual size dimorphism and movement patterns have been well studied (e.g., McFarlane and Beamish 1990; Kimura et al. 1998). However, few studies have examined the potential linkage between sex-biased growth and dispersal in marine fishes.

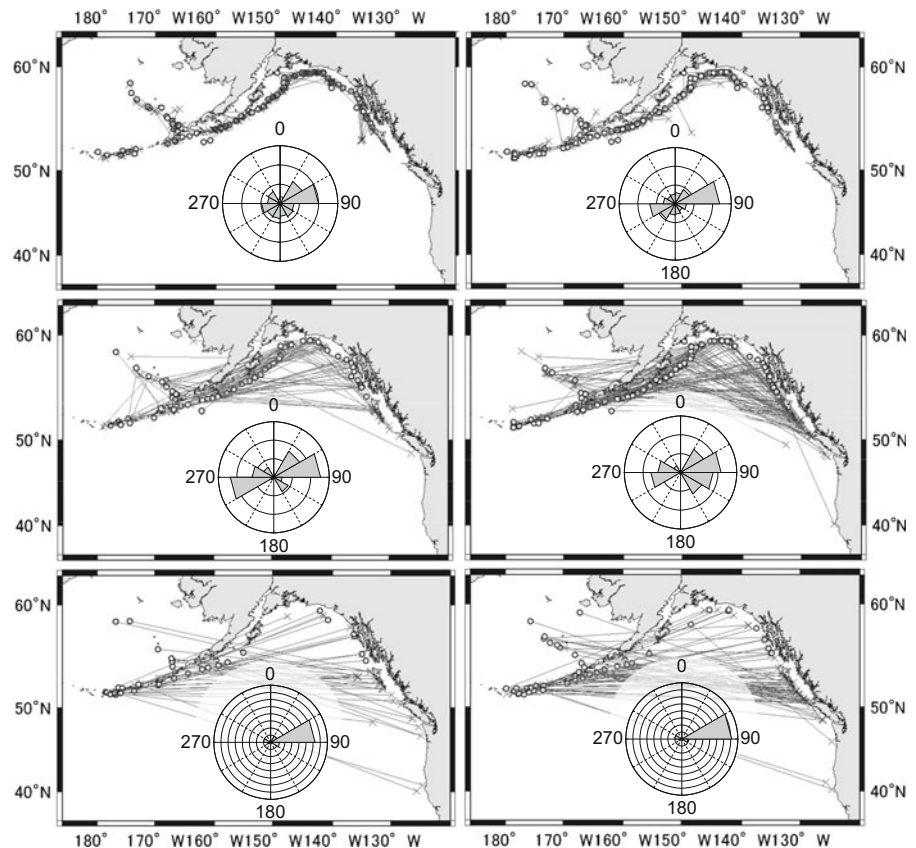
The sablefish is a popular commercially important groundfish species occurring at depths of 200–1500 m from Baja California along the Pacific coast, through the Gulf of Alaska and the Aleutian Islands, along the Russian coast, Kamchatka Peninsula, and southern coast of Japan (Allen and Smith 1988). Sablefish migrate horizontally (Sasaki 1985; McFarlane et al. 1997) and they use various depths during their life stages; for example, they spawn at 300–500 m, juveniles occupy relatively shallow waters (<200 m), and adults occupy depths greater than 200 m (Beamish et al. 2008). Sablefish have a very long life expectancy; the oldest recorded age, estimated from otoliths, was 114 years (Beamish and McFarlane 2000). Furthermore, sablefish grow rapid-

ly until age 5, when they begin to mature; after maturity, they grow slowly, but males reach in excess of 70 cm and females reach 90–100 cm (King et al. 2001). Many studies have examined the movement and growth of tagged sablefish (Heifetz and Fujioka 1991; Heifetz et al. 1998; Kimura et al. 1998; Maloney and Sigler 2008), but few studies have demonstrated sex-biased dispersal in relation to growth. In this study, we examined the sex-biased dispersal and growth of sablefish and the relationship between movement and growth using long-term tagging data.

## Materials and methods

Tagging was conducted by teams from Japan (National Research Institute of Far Seas Fisheries, Fisheries Research Agency) and the United States (Alaska Fisheries Science Center). From June 1980 to September 1984, a total of 58,918 sablefish were captured using longline gear (the average depth of the fishing points ranged from 183 to 800 m, with a minimum of 40 m and maximum of 1270 m) in the Bering Sea, Aleutian Region, and Gulf of Alaska (Fig. 1). The point of capture, date of capture, and fork length were recorded for each fish, and each individual was tagged with an anchor tag (Floy FD-68) and released. The vinyl tubing on each tag was 2 mm in diameter and 60 mm long. The mean fork length  $\pm$  S.D. of released fish was  $59 \pm 7.1$  cm, which exceeds the average size at maturity of this species (Mason et al. 1983). Between 1980 and 2007, 3319 sablefish were reported recaptured by fishermen from the Bering Sea to the northeastern Pacific. Since the recaptures were made by fishermen, the sampling area is restricted to fishing areas only, with restricted ranges in depth. For each recaptured fish, the fishermen reported the fishing location, date, sex, fork length, and tag number, although complete data were not always provided. Of the 3319 sablefish reported to have been recaptured between 1980 and 2007, 350 were males, 572 were females, and the sex was unknown for the remaining 2397 fish. Of the 922 sexed fish, we analyzed data for 659 (259 males, 400 females) individuals for which there were reliable data on recapture location, fork length at the time of release and recapture, and elapsed time longer than 10 days. Straight movement distances (km) and

**Fig. 1** Maps showing the locations of release (○) and recapture (×) sites of male (left panels) and female (right panels) sablefish for different movement distances (top, <500 km; middle, 500–2000 km; bottom, >2000 km) between 1980 and 2007. The rose diagram indicates the frequency distribution of migratory direction (°) by percentage (one division=10%)



migratory direction (°) were calculated from the latitudes and longitudes of the release and recapture points using an azimuthal equidistant projection method. The growth increment (cm) and elapsed time (days) were calculated for each recaptured fish. We assumed that the effects of tagging on growth and dispersal are independent of sex.

The growth increment,  $\Delta L$ , was dependent on the initial fork length and elapsed time (Quinn and Deriso 1999). To evaluate the growth of individuals, we calculated growth increment deviations using the von Bertalanffy model. The von Bertalanffy growth model for individual-based tag-recapture data can be described as (Quinn and Deriso 1999):  $L_{2,i} = L_{\infty}[1 - \exp(-K \cdot t_i)] + L_{1,i} \cdot \exp(-K \cdot t_i) + \varepsilon_i$ , where  $L_{1,i}$  and  $L_{2,i}$  are the length measurements for the  $i^{\text{th}}$  fish at release and recapture, respectively,  $L_{\infty}$  is the asymptotic length,  $K$  is the rate by which the growth curve approaches the asymptote,  $t_i$  is the elapsed time for the  $i^{\text{th}}$  fish, and  $\varepsilon_i$  is an individual variability term with mean zero and variance  $(\sigma + \nu \cdot \Delta L)^2$ . The model parameters were estimated by the maximum

likelihood method using the Solver maximization algorithms in Microsoft Excel. Then, growth increment deviations were calculated as the differences between the observed and expected growth increment values,  $\ln(\Delta L) - \ln(\Delta \hat{L})$ . Sex-specific values of  $L_{\infty}$  and  $K$  were also calculated, and the Akaike information criterion (AIC) was used to evaluate sexual differences in model parameters. Four models,  $M_1$ – $M_4$ , were fitted to the data, each representing a different hypothesis about the  $L_{\infty}$  and  $K$  values (Table 2).

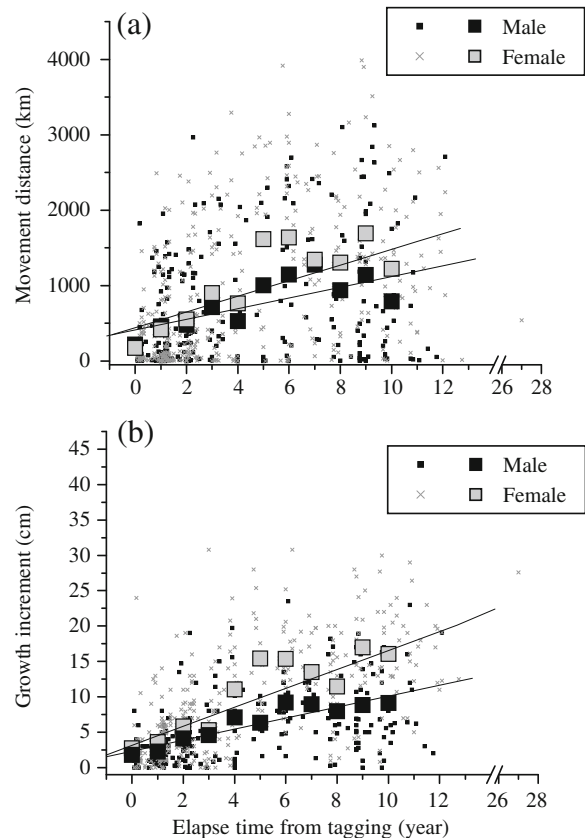
## Results

The mean fork lengths  $\pm$  S.D. at release of recaptured fish were  $57 \pm 5.8$  cm for males and  $60 \pm 6.7$  cm for females. The mean fork lengths  $\pm$  S.D. at recapture were  $63 \pm 6.3$  cm for males and  $69 \pm 8.2$  cm for females. The individual with the longest elapsed time (27 years) grew 27 cm between release and recapture. Females appeared to be recaptured more commonly

over the years, probably due to size-dependent commercial fishing. The migratory direction of the tag-recaptured sablefish was most commonly a north to east course for both sexes, and this pattern was more pronounced with increasing movement distance (Fig. 1). There were no significant differences in the migratory direction between the sexes (Kolmogorov-Smirnov test,  $p=0.74$ ).

Despite the lack of significant differences in elapsed times between the sexes, both the straight movement distance and the growth increment differed significantly between the sexes (Table 1). The straight movement distance was significantly longer for females than for males; the mean values were 712 km for males and 877 km for females. The longest straight movement distances were 3123 km for a male and 3987 km for a female, both of which were recaptured 9 years after release. The growth increments were larger for females than for males; the mean growth increment was 5.6 cm for males and 8.8 cm for females. In addition, females grew larger and moved longer distances than males after controlling for elapsed time (Fig. 2). Analysis of covariance identified significant differences in the regression lines for both growth increment and movement distance on elapsed time between the sexes (growth increment:  $F_{1,655}=24.7$ ,  $p<0.001$ , movement distance:  $F_{1,655}=7.06$ ,  $p=0.008$ ).

The growth increment deviations obtained from von Bertalanffy growth models increased slightly but significantly with increasing distance moved per elapsed time (Fig. 3;  $r=0.177$ ,  $p<0.001$ ). However, the relationship between the distance moved per elapsed time and the growth increment deviation differed between the sexes. Analysis of covariance did not identify a significant difference in regression slopes between sexes, but did identify a significant difference in growth increment deviations between sexes after controlling for the distance moved per

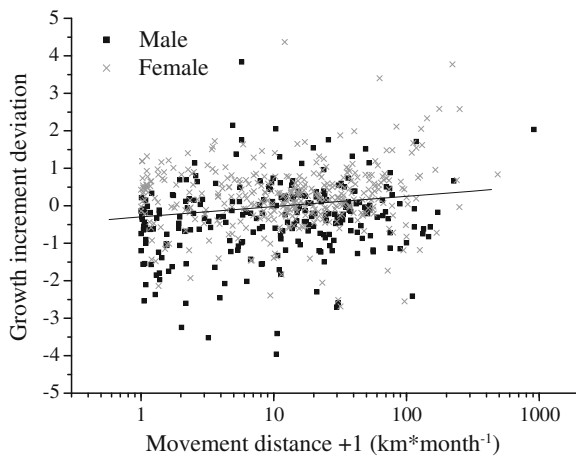


**Fig. 2** Relationships between (a) elapsed time and movement distance and (b) growth increment for male and female sablefish. Small symbols indicate individual data and large symbols indicate mean values for each elapsed time class. The regression lines in (a) male movement distance =  $70.86 \text{ elapsed time} + 410.27$ ,  $r=0.31$ , female movement distance =  $103.67 \text{ elapsed time} + 443.27$ ,  $r=0.41$  and (b) male growth increment =  $0.78 \text{ elapsed time} + 2.32$ ,  $r=0.52$ , female growth increment =  $1.34 \text{ elapsed time} + 3.16$ ,  $r=0.67$

elapsed time (comparison of slopes:  $F_{1,627}=1.54$ ,  $p=0.215$ , comparison of adjusted means:  $F_{1,628}=65.1$ ,  $p<0.001$ ). In addition, the estimated von Bertalanffy growth parameters ( $L_{\infty}$  and  $K$ ) varied between the

**Table 1** Mean $\pm$ S.D. ( $n$ ) elapsed time, movement distance, and growth increment for tag-recaptured sablefish in the North Pacific

|                             | Males                                    | Females                                  | Mann-Whitney $U$ -test |
|-----------------------------|------------------------------------------|------------------------------------------|------------------------|
| Elapsed time                | 4.3 $\pm$ 3.48 years (259)               | 4.2 $\pm$ 3.67 years (400)               | $p=0.770$              |
| Movement distance           | 712 $\pm$ 800 km (259)                   | 877 $\pm$ 920 km (400)                   | $p=0.027$              |
| Movement distance per month | 27.6 $\pm$ 66.0 km $\text{m}^{-1}$ (259) | 29.0 $\pm$ 45.8 km $\text{m}^{-1}$ (400) | $p=0.070$              |
| Growth increment            | 5.64 $\pm$ 5.14 cm (259)                 | 8.76 $\pm$ 7.29 cm (400)                 | $p<0.001$              |
| Annual growth increment     | 2.54 $\pm$ 6.5 cm $\text{yr}^{-1}$ (259) | 3.53 $\pm$ 8.6 cm $\text{yr}^{-1}$ (400) | $p<0.001$              |



**Fig. 3** Relationship between distance moved per month and growth increment deviation calculated as  $\ln(\Delta L/\Delta \hat{L})$ , where  $\Delta L$  is the observed growth increment and  $\Delta \hat{L}$  is the expected growth increment obtained from the von Bertalanffy growth models  $M_1$  (see Table 1), for male and female sablefish. Regression line, growth increment deviation =  $0.28 \log(\text{movement distance} + 1) - 0.30$ ,  $r = 0.18$

sexes. The model that assumed that  $L_\infty$  was sex-specific had the smallest AIC among the four possible models and was therefore the best model considered here (Table 2).

## Discussion

From the analysis of long-term tagging data, we found that female sablefish grew larger and moved longer distances than male sablefish. In addition, there was a weak but significant positive correlation between movement distance and growth for sablefish. Therefore, sex-biased dispersal may be a proximate factor responsible for the sex-biased growth in sablefish.

**Table 2** Von Bertalanffy growth models  $M_1$ – $M_4$  fitted to tag-recaptured sablefish, with estimated parameters ( $L_\infty$ , asymptotic length;  $K$ , growth coefficient), number of parameters ( $n$ ) and

| Model |          | $L_\infty$ (cm) |        |          | $K$ (year <sup>-1</sup> ) |        | $n$ | $\Delta\text{AIC}$ |
|-------|----------|-----------------|--------|----------|---------------------------|--------|-----|--------------------|
|       |          | Male            | Female |          | Male                      | Female |     |                    |
| $M_1$ | Constant | 86.1            | 86.1   | Constant | 0.08                      | 0.08   | 4   | 127.5              |
| $M_2$ | Variable | 73.5            | 86.5   | Constant | 0.11                      | 0.11   | 5   | 0                  |
| $M_3$ | Constant | 83.8            | 83.8   | Variable | 0.06                      | 0.13   | 5   | 10.7               |
| $M_4$ | Variable | 74.5            | 85.6   | Variable | 0.10                      | 0.11   | 6   | 1.3                |

Consistent with our results, many reports have shown that females grow larger than males in demersal fishes, e.g., rockfishes (*Sebastes spp.*; Love et al. 2002), Japanese flounder (*Paralichthys olivaceus*; Udagawa and Fukuda 2008), pointhead flounder (*Hippoglossoides pinetorum*; Itaya and Fujioka 2006), brown sole (*Pleuronectes herzensteini*; Yabuki 2008), threadfin hakeling (*Laemonema longipis*; Narimatsu et al. 2008), walleye pollock (*Theragra chalcogramma*; Nishimura et al. 2001), pacific grenadier (*Coryphaenoides acrolepis*; Andrews et al. 1999), yellow goosfish (*Lophius litulon*; Ito et al. 2008), and sablefish (McFarlane and Beamish 1990). Although the proximate factors driving the sexual size dimorphism are unclear (e.g., sexual niche segregation), the ultimate factor driving female-biased size dimorphism in demersal fishes can be hypothesized. It is expected that sexual selection acting upon male body size would not operate well in most demersal fishes, including sablefish, because of promiscuous mating and external fertilization in dark environments (cf. Stearns 1992). By contrast, the fecundity of fishes increases isometrically with body weight (Roff 1992); therefore, fecundity selection should operate intensely in fishes. In female sablefish, the annual production of eggs increases with body size, from 180,000–280,000 eggs for 65–75 cm up to one million for 90–100 cm (King et al. 2001). Consequently, females should grow larger than males to produce more eggs. Therefore, fecundity selection would provide a stronger force than sexual selection in demersal fishes, including sablefish, resulting in a female-biased sexual size dimorphism.

Sex-biased dispersal has been reported for several fishes (Hendry et al. 2004; Consuegra and Garcia de Leaniz 2007). Tag-recapture studies found that some demersal fishes could show seasonal and local

Akaike information criterion relative to the best-fit model ( $\Delta\text{AIC}$ ). Constant, male = female; variable, male  $\neq$  female

movement (Hunt et al. 1999; McDermott et al. 2005), but little is known about the sex-biased dispersal patterns in fishes. Beamish and McFarlane (1988) indicated that adult sablefish show little movement behavior, but our long-term data indicate that adult sablefish undergo long-distance movement. In the great white shark and brook trout, males moved longer distances than females (Pardini et al. 2001; Hutchings and Gerber 2002). However, our sablefish results conflict with these previous findings; we found that female sablefish dispersed longer distances than males. Conversely, among migratory salmonids, a greater number of females migrate to productive habitats and grow to larger sizes compared to males (Jonsson and Jonsson 1993; Koizumi et al. 2006). Kimley (1987) showed that female hammerhead sharks moved offshore at a smaller size than males, and this offshore migration results in female-biased rapid growth, as inferred from larger stomach content masses in females than in males of the same size. Analogously, female sablefish might perform more or larger feeding migrations compared to males. This idea is supported by the fact that the growth increment deviation increased with increasing movement distance per year.

In this study, movement distance was determined as a straight distance between the points of capture and recapture; however, the actual movement is probably along the continental shelf and is likely more complex. Investigation of actual movement routes may elucidate the relationships between growth and movement distance. In recent years, the use of archival tags has made it possible to examine specific movements of marine fishes, e.g., the North Sea cod (*Gadus morhua*; Turner et al. 2002) and the Atlantic blue-fin tuna (*Thunnus thynnus*; Sibert et al. 2006). In addition, the effects of other environmental factors (e.g., reproductive state, condition factors, and temporal variation) on dispersal and growth were not addressed in this study, but should not be ruled out (cf. Pettersson et al. 2004). Female-biased dispersal is hypothesized to occur in sablefish, but much about this remains unknown and merits further investigation.

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## References

- Allen MJ, Smith GB (1988) Atlas and zoogeography of common fisheries in the Bering Sea and northeastern Pacific. U. S. Department of Commerce, NOAA Technical Reports. NMFS 66:1–151
- Andrews AH, Cailliet GM, Coale KH (1999) Age and growth of the Pacific grenadier (*Coryphaenoides acrolepis*) with age estimate validation using an improved radiometric ageing technique. *Can J Fish Aquat Sci* 56:1339–1350
- Beamish RJ, McFarlane GA (1988) Resident and dispersal behavior of adult sablefish (*Anoplopoma fimbria*) in the slope waters off Canada's west coast. *Can J Fish Aquat Sci* 45:152–164
- Beamish RJ, McFarlane GA (2000) Reevaluation of the interpretation of annuli from otoliths of a long-lived fish, *Anoplopoma fimbria*. *Fish Res* 46:105–111
- Beamish RJ, King JR, McFarlane GA (2008) Country Report, Canada. In: Beamish RJ (ed) Impacts of Climate and Climate Change on the Key Species in the Fisheries in the North Pacific, pp 15–55, PICES Sci. Rep. No. 35, 217
- Beacham TD, Murray CB (1985) Variation in length and body depth of pink salmon (*Oncorhynchus gorbuscha*) and chum salmon (*O. keta*) in southern British Columbia. *Can J Fish Aquat Sci* 42:312–319
- Blair GR, Rogers DE, Quinn TP (1993) Variation in life history characteristics and morphology of sockeye salmon in the Kvichak River system, Bristol Bay, Alaska. *Trans Am Fish Soc* 122:550–559
- Clarke AL, Sæther BE, Røskoft E (1997) Sex biases in avian dispersal: a reappraisal. *Oikos* 79:429–438
- Consuegra S, Garcia de Leaniz C (2007) Fluctuating sex ratios, but no sex-biased dispersal, in a promiscuous fish. *Evol Ecol* 21:229–245
- Dobson FS (1982) Competition for mates and predominant juvenile male dispersal in mammals. *Anim Behav* 30:1183–1192
- Greenwood PJ (1980) Mating systems, philopatry and dispersal in birds and mammals. *Anim Behav* 28:1140–1162
- Greenwood PJ (1983) Mating systems and the evolutionary consequences of dispersal. In: Swingland IR, Greenwood PJ (eds) The ecology of animal movement. Clarendon, Oxford, pp 116–131
- Heifetz J, Fujioka T (1991) Movement dynamics of tagged sablefish in the northeastern Pacific. *Fish Res* 11:355–374
- Heifetz J, Anderl D, Maloney NE, Rutecki TL (1998) Age validation and analysis of ageing error from marked and recaptured sablefish, *Anoplopoma fimbria*. *Fish Bull* 97:256–263
- Hendry AP, Castric V, Kinnison MT, Quinn TP (2004) The evolution of philopatry and dispersal. In Hendry AP and Stearns SC (ed) Evolution Illuminated. Salmon and their Relatives, Oxford University Press, pp 52–91
- Holtby LB, Healey MC (1990) Sex-specific life history tactics and risk-taking Coho salmon. *Ecology* 71:678–690
- Hunt JJ, Stobo WT, Almeida F (1999) Movement of Atlantic cod, *Gadus morhua*, tagged in the Gulf of Maine area. *Fish Bull* 97:842–860
- Hutchings JA, Gerber L (2002) Sex-biased dispersal in a salmonid fish. *Proc R Soc Lond B* 269:2487–2493

- Itaya K, Fujioka T (2006) The growth of pointhead flounder *Hippoglossoides pinetorum* in Ishikari Bay. Hokkaido Fisheries Sci Rep 70:89–94 (In Japanese)
- Ito M, Hattori T, Narimatsu Y, Ueda Y (2008) Stock assessment and evaluation for yellow goosefish *Lophius litulon*. In: Marine Fisheries Stock Assessment and Evaluation for Japanese Waters, Fisheries Agency and Fisheries Research Agency of Japan, pp 774–788
- Jonsson B, Jonsson N (1993) Partial migration: niche shift versus sexual maturation in fishes. Rev Fish Biol Fish 3:348–365
- Katano O (1998) Growth of dark chub, *Zacco temminckii* (Cyprinidae), with a discussion of sexual size differences. Environ Biol Fish 52:305–312
- Kimley AP (1987) The determinants of sexual segregation in the scalloped hammerhead shark, *Sphyrna lewini*. Environ Biol Fish 18:27–40
- Kimura DK, Shimada AM, Shaw FR (1998) Stock structure and movement of tagged sablefish, *Anoplopoma fimbria*, in offshore northeast Pacific waters and the effects of El Niño-Southern Oscillation on migration and growth. Fish Bull 96:462–481
- King JR, McFarlane GA, Beamish RJ (2001) Incorporating the dynamics of marine systems into the stock assessment and management of sablefish. Prog Oceanogr 49:619–639
- Koizumi I, Yamamoto S, Maekawa K (2006) Female-biased migration of stream-dwelling Dolly Varden in Shiisorapuchi River, Hokkaido, Japan. J Fish Biol 68:1513–1529
- Love MS, Yoklavich M, Thorsteinson L (2002) The Rockfishes of the Northeast Pacific. University of California Press
- Maloney NE, Sigler MF (2008) Age-specific movement patterns of sablefish (*Anoplopoma fimbria*) in Alaska. Fish Bull 106:305–316
- Mason JC, Beamish RJ, McFarlane GA (1983) Sexual maturity, fecundity, spawning, and early life history of sablefish (*Anoplopoma fimbria*) off the Pacific coast of Canada. Can J Fish Aquat Sci 40:2126–2134
- McDermott SF, Fritz LW, Haist AV (2005) Estimating movement and abundance of Atka mackerel (*Pleurogrammus monopterygius*) with tag-release-recapture data. Fish Oceanogr 14(suppl 1):113–130
- McFarlane GA, Beamish R (1990) Effect of an external tag on growth of sablefish (*Anoplopoma fimbria*), and consequences to mortality and age at maturity. Can J Fish Aquat Sci 47:1551–1557
- McFarlane GA, Saunders MW, Thomson RE, Perry RI (1997) Distribution and abundance of larval sablefish, (*Anoplopoma fimbria*), off the west coast of Vancouver Island, and linkage to physical oceanography. In: Saunders M, Wilkens M (ed) Biology and Management of sablefish, *Anoplopoma fimbria*, pp 27–38. U.S. Dep. Commer., NOAA Tech. Rep. NMFS 130
- Narimatsu Y, Ito M, Hattori T, Ueda Y (2008) Stock assessment and evaluation for threadfin hakeling *Laemonema longipis*. In: Marine fisheries stock assessment and evaluation for Japanese waters, Fisheries Agency and Fisheries Research Agency of Japan, pp 718–728
- Nishimura A, Yanagimoto T, Mito K, Katakura S (2001) Interannual variability in growth of walleye Pollock, *Theragra chalcogramma* in the central Bering Sea. Fish Oceanogr 10:367–375
- Pardini AT, Jones CS, Noble LR, Kreiser B, Malcolm H, Bruse BD, Stevens JD, Cliff G, Scholl MC, Francis M, Duffy CAJ, Martin AP (2001) Sex-biased dispersal of great white sharks. Nature 412:139–140
- Pettersson LB, Ramnarine IW, Becher SA, Mahabir R, Magurran AE (2004) Sex ratio dynamics and fluctuating selection pressures in natural populations of the Trinidadian guppy, *Poecilia reticulata*. Ecol Sociobiol 55:461–468
- Quinn II TJ, Deriso RB (1999) Quantitative fish dynamics. Oxford University Press
- Quinn TP, Hendry AP, Buck GB (2001) Balancing natural and sexual selection in sockeye salmon: interactions between body size, reproductive opportunity and vulnerability to predation by bears. Evol Ecol Res 3:917–937
- Roff DA (1992) The evolution of life histories. Chapman and Hall, New York
- Ruckstuhl KE, Neuhaus P (2002) Sexual segregation in ungulates: a comparative test of three hypotheses. Biol Rev 77:77–96
- Sasaki T (1985) Studies on the sablefish resources in the North Pacific Ocean. F Bull Far Seas Fish Res Lab 22:1–108
- Sibert JR, Lutcavage ME, Nielsen A, Brill RW, Wilson SG (2006) Interannual variation in large-scale movement of Atlantic bluefin tuna (*Thunnus thynnus*) determined from pop-up satellite archival tags. Can J Fish Aquat Sci 63:2154–2166
- Stearns SC (1992) The evolution of life histories. Oxford University Press
- Tamate T, Maekawa K (2004) Female-biased mortality rate and sexual size dimorphism of migratory masu salmon, *Oncorhynchus masou*. Ecol Freshw Fish 13:96–103
- Tamate T, Maekawa K (2006) Latitudinal variation in sexual size dimorphism of sea-run masu salmon, *Oncorhynchus masou*. Evolution 60:196–201
- Turner K, Righton D, Metcalfe JD (2002) The dispersal patterns and behavior of North Sea cod (*Gadus morhua*) studied using electronic data storage tags. Hydrobiologia 483:201–208
- Udagawa T, Fukuda M (2008) Stock assessment and evaluation for Japanese flounder *Paralichthys olivaceus*. In: Marine fisheries stock assessment and evaluation for Japanese waters, Fisheries Agency and Fisheries Research Agency of Japan, pp 1163–1175
- Yabuki K (2008) Stock assessment and evaluation for brown sole *Pleuronectes herzensteini*. In: Marine fisheries stock assessment and evaluation for Japanese waters, Fisheries Agency and Fisheries Research Agency of Japan, pp 1356–1367